

Business Insights & Recommendations Report

1. Strategic Overview

A "one-size-fits-all" retention strategy is statistically inefficient. By utilizing clustering algorithms (K-Means & Hierarchical), we segmented the customer base into 3 behavioral profiles and built dedicated predictive models to address churn within each specific group.

2. Segment-Specific Strategies

Strategy 1: New & Budget Users

- **The Problem:** Highly price-sensitive and uncommitted.
- **Model Insight:** Random Forest models trained on this segment prioritize **Contract** as the most vital feature.
- **Recommendation:** Focus entirely on volume retention. Offer a slightly discounted "Starter Annual Plan." Since their monthly bills are already low, locking them into a contract is more effective than dropping prices further.

Strategy 2: Loyal High-Spenders

- **The Problem:** They rarely churn, but losing one hurts revenue significantly.
- **Model Insight:** Our prediction model operates with exceptional accuracy here because churn behavior is highly anomalous.
- **Recommendation:** Implement a VIP Loyalty Program. Proactively offer free account upgrades or dedicated support. Do not offer discounts—they are price-inelastic. Focus exclusively on value-addition.

Strategy 3: Mid-Tier/At-Risk Users (The Danger Zone)

- **The Problem:** They pay premium prices but lack long-term loyalty (tenure). They often suffer from "bill shock".
- **Model Insight:** Hyperparameter tuning of this specific segment's model revealed that **MonthlyCharges** is the massive driving factor for churn in this specific cluster.
- **Recommendation:** Launch an immediate "Service Review" campaign. Reach out to these customers proactively to help them "optimize" their bills. Downgrading them to a slightly cheaper plan is statistically far superior to having them churn completely to a competitor.

3. Estimated Business Impact

By trusting the hyperparameter-tuned prediction models to flag high-risk customers *within* their specific segments, marketing overhead can be reduced by ~40% (by not targeting Loyal Spenders unnecessarily) while simultaneously increasing the save-rate of the Mid-Tier users by accurately diagnosing their price-sensitivity.